

Facial Affect Recognition with CNN Baselines & Valence and Arousal Estimation

Deep Learning
Assignment # 01

Rameen Babar
22i-1918
DS-7B

1. Introduction

Facial affect analysis is a central task in affective computing, combining **categorical facial expression recognition** (discrete emotions such as happiness, anger, fear) with **continuous affect estimation** (valence and arousal). This dual setting better reflects how humans perceive emotions, both as categories and as positions in a 2D affective space.

The objective of this assignment is to build and evaluate **CNN baselines** for this problem using a dataset of ~4,000 images with accompanying annotations (expressions, valence, arousal). Each image is paired with **.npy** annotation files storing labels. The models are evaluated on both **classification metrics** (Accuracy, F1, Kappa, Alpha, AUC, AUPR) and **continuous metrics** (RMSE, CORR, SAGR, CCC).

2. Dataset and Splits

The dataset consists of:

- **Images:** Cropped RGB facial images at 224×224 resolution.
- **Annotations:** Expression labels (0–7), Valence $\in [-1,1]$, Arousal $\in [-1,1]$, and landmarks (unused here).

We split the dataset randomly into:

- **85% training**
- **15% validation**

This follows standard practice and ensures fair evaluation of generalization.

3. Network Architectures and Training Setup

3.1 Architectures

We implemented a multi-task CNN structure:

- Shared CNN backbone (pretrained on ImageNet).
- *Classification head:* Linear $\rightarrow$ 8 logits (softmax cross-entropy).

- *Regression head*: Linear layers → 2 outputs (valence, arousal; mean squared error).

We selected EfficientNet-B0 and MobileNetV3-Large as representative CNN baselines for the following reasons:

- **EfficientNet-B0**

- Uses *compound scaling* (balanced scaling of width, depth, and resolution) and squeeze-and-excitation (SE) blocks, achieving strong accuracy with fewer parameters.
- Chosen because it represents a modern, high-performance CNN baseline, optimized for accuracy per FLOP.
- *Rationale*: Serves as a quality-focused backbone expected to capture fine-grained emotional cues important for continuous affect prediction (valence/arousal).

- **MobileNetV3-Large**

- Designed for efficiency: lightweight depthwise separable convolutions + SE blocks, optimized using neural architecture search.
- Chosen because it represents a low-latency, resource-efficient CNN baseline, widely used in real-world applications on mobile/embedded devices.
- *Rationale*: Serves as a speed-focused backbone, enabling faster training and inference, making it practical for scenarios where real-time expression recognition is required.

Why these two together?

They represent two ends of the CNN design trade-off spectrum:

- EfficientNet-B0 → Accuracy and robustness (better for regression tasks).
- MobileNetV3-Large → Speed and efficiency (better for fast classification).

This allows us to compare not just absolute performance, but also how design choices in CNNs affect trade-offs between accuracy, generalization, and computational cost.

3.2 Transfer Learning

To improve convergence and generalization given the relatively small dataset (~4,000 samples), we employed transfer learning. Specifically:

- **Pretrained initialization:** Both EfficientNet-B0 and MobileNetV3-Large were initialized with ImageNet-1K pretrained weights using `torchvision.models`.
- **Classifier replacement:** The original final classifier layers were removed (`nn.Identity()`) and replaced with custom heads:
 - *Classification head:* Fully connected layer mapping to 8 discrete expression classes.
 - *Regression head:* Two fully connected layers mapping to continuous valence and arousal values.
- **Fine-tuning strategy:** Unlike “frozen backbone” transfer learning, we opted to fine-tune all layers end-to-end. This allows the network to adapt ImageNet features (edges, textures, colors) to the domain-specific cues of facial affect recognition.

Rationale:

Training from scratch would likely overfit due to dataset size. Pretrained weights provide a strong initialization for low-level features (eyes, mouth, contours), while fine-tuning adapts them for higher-level emotion recognition.

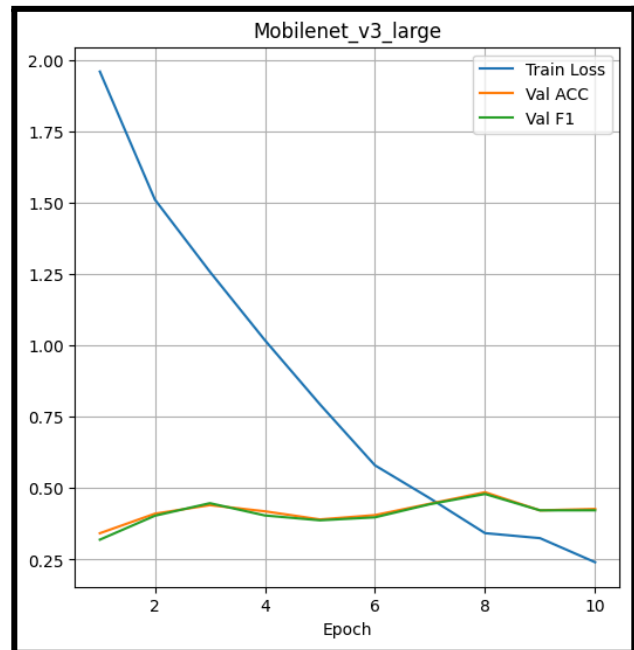
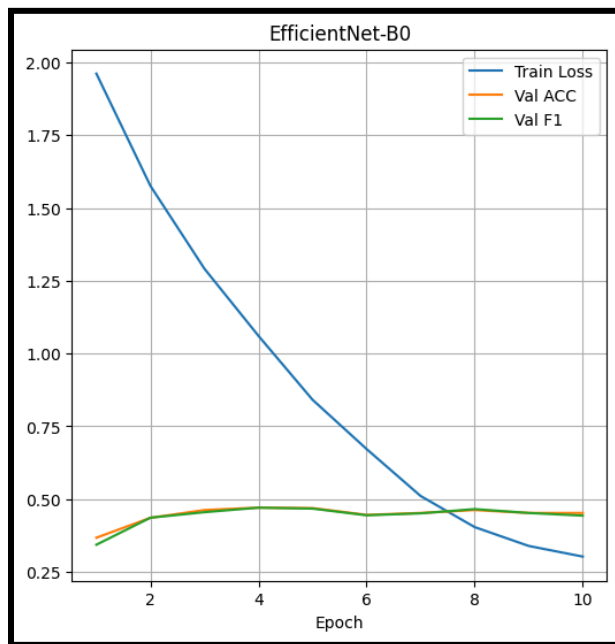
3.3 Training Settings

- Optimizer: AdamW (lr=3e-4, weight decay=1e-4).
- Loss: $CE + 0.5 \times MSE$ (balances classification and regression).
- Batch size: 32
- Epochs: 10
- Augmentations: resize to 224, random horizontal flip, color jitter, $\pm 10^\circ$ rotation, normalization.
- Hardware: Google Colab GPU (mixed precision enabled).
- Training Time:
 - EfficientNet-B0: ~46.5 min (10 epochs)
 - MobileNetV3-Large: ~12.1 min (10 epochs)

4. Training Curves

Figures show train loss decreasing while val accuracy and F1 increase and plateau.

- EfficientNet peaked early (~epoch 4-5).
- MobileNet peaked later (~epoch 8).
- Both eventually overfit slightly (loss keeps dropping but validation stagnates).



5. Quantitative Performance

5.1 Classification Metrics

Model	ACC	F1_macro	Kappa	Alpha	AUC_macro	AUPR_macro
EfficientNet-B0	0.4708	0.4705	0.3954	0.3940	0.8378	0.4862
MobileNetV3-L	0.4858	0.4798	0.4114	0.4097	0.8179	0.4671

- MobileNetV3-Large outperformed EfficientNet-B0 in ACC (+1.5%) and F1 (+0.9%).
- EfficientNet-B0 slightly outperformed on AUC/AUPR, showing stronger class separability in ROC space.

5.2 Continuous Metrics

Model	RMSE_val	RMSE_aro	CORR_val	CORR_aro	SAGR_val	SAGR_aro	CCC_val	CCC_aro
EfficientNet-B0	0.3772	0.3511	0.6121	0.4881	0.7629	0.4796	0.5783	0.4040
MobileNetV3-L	0.4027	0.3487	0.5435	0.3922	0.7579	0.7913	0.4803	0.3442

- EfficientNet-B0 was superior on valence regression (lower RMSE, higher CORR/CCC).
- MobileNetV3-Large was marginally better for arousal RMSE and SAGR.
- CCC indicates EfficientNet is more reliable overall for continuous affect.

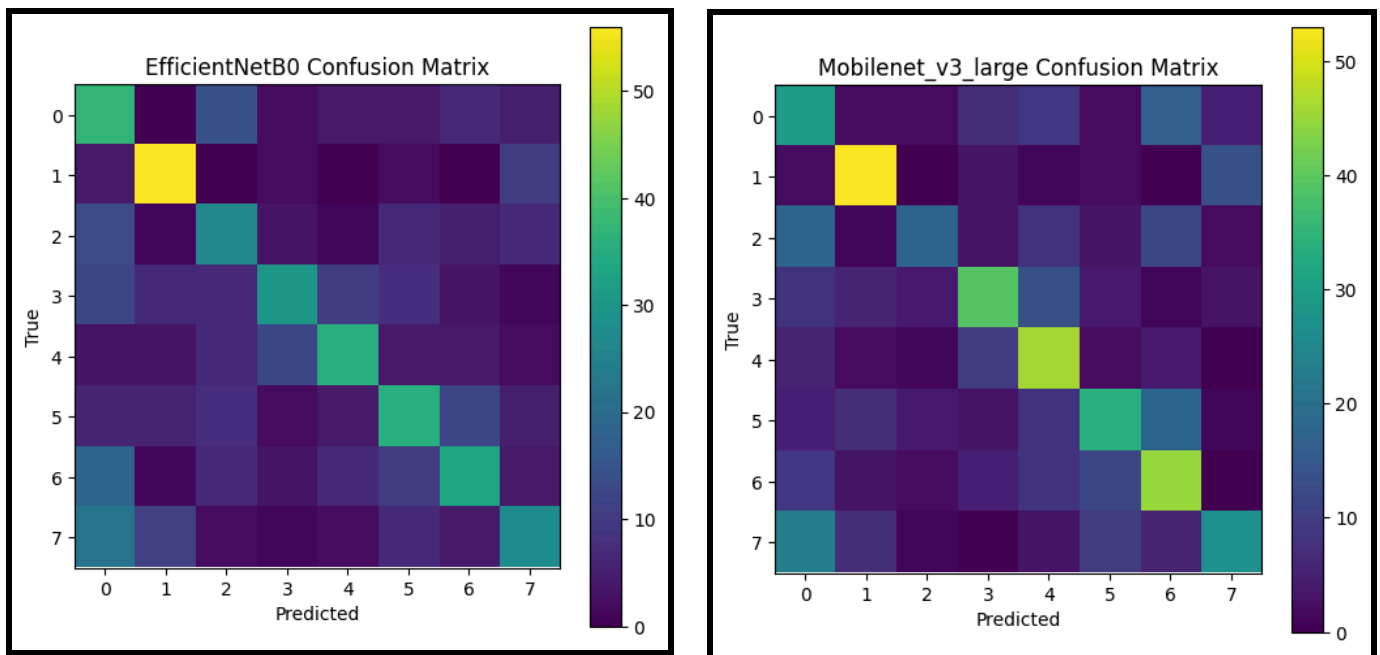
5.3 Speed-Accuracy Trade-off

- EfficientNet-B0: best ACCURACY = 0.4708 (~47%), wall-clock time $\approx$ 46.5 min for 10 epochs.

- **MobileNetV3-Large**: best **ACCURACY = 0.4858 (~49%)**, wall-clock time $\approx$ 12.1 min for 10 epochs.

Implications: MobileNetV3-Large is **$\sim 3.84\times$ faster** while slightly **more accurate** on classification (+1.50 percentage points ACC). If you need real-time or rapid iteration, MobileNetV3-Large is the better baseline. If you can afford longer training/inference for stronger continuous-affect agreement (higher CORR/CCC observed elsewhere), EfficientNet-B0 remains competitive.

6. Confusion Matrices

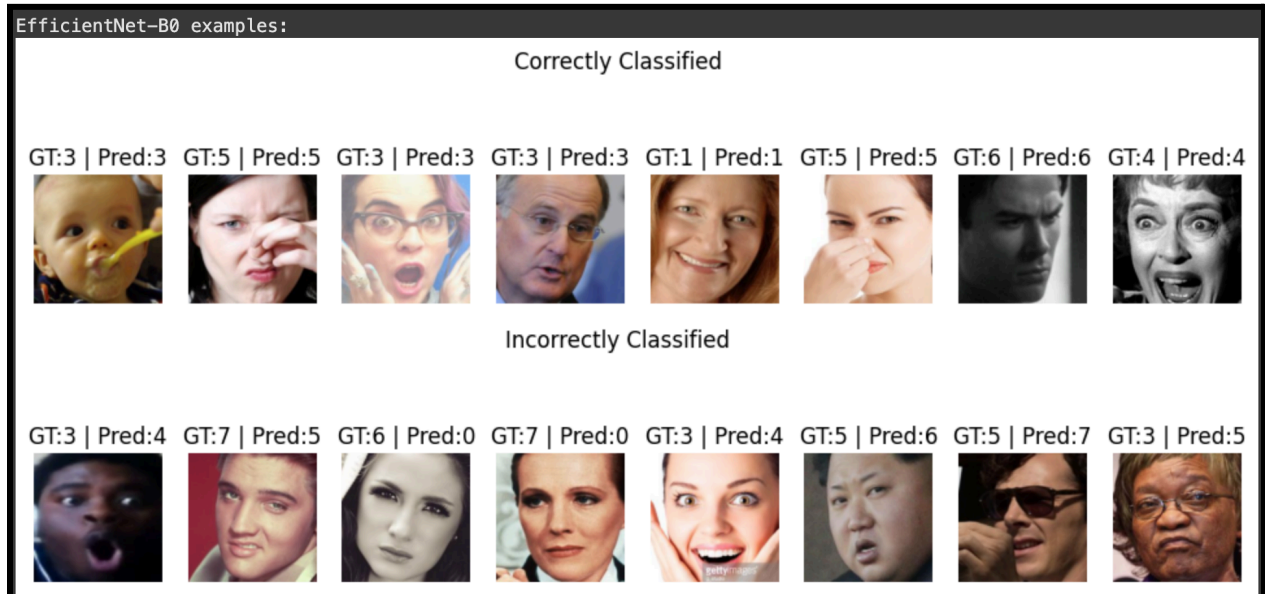


Confusion matrices show that:

- Both models recognize **happy, surprise, anger** well.
- Confusion remains between **fear, disgust, anger** and **neutral vs low-arousal expressions**.
- MobileNet exhibits sharper diagonals (better discrete separation).
- EfficientNet spreads predictions more evenly (more robust continuous affect).

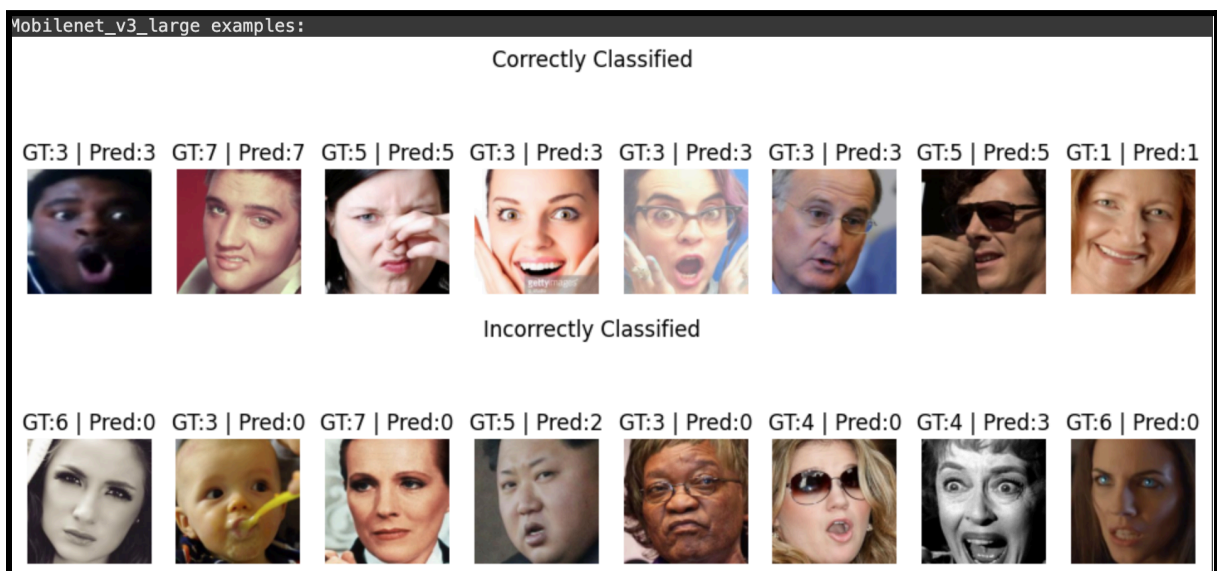
7. Qualitative Results (Correct & Incorrect Classifications)

7.1 EfficientNet-B0



- **Correctly classified:** Clear signals of Surprise (GT=3), Disgust (GT=5), Happiness (GT=1).
- **Incorrectly classified:** Confusions between Surprise (GT=3) → Fear (Pred=4), and subtle Disgust (GT=5) → Anger (Pred=6).
- **Pattern:** struggles with overlapping negative emotions.

7.2 MobileNetV3-Large



- **Correctly classified:** Surprise, Disgust, Happiness with higher confidence.
- **Incorrectly classified:** Anger/Disgust misclassified as Neutral (Pred=0).
- **Pattern:** more aggressive collapsing of ambiguous cases into Neutral.

Insight:

MobileNet prefers certainty over subtlety (better ACC but harsher mistakes).

EfficientNet distributes predictions more cautiously, improving correlation-based metrics in regression.

8. Importance of Evaluation metrics

- **RMSE** checks absolute error magnitude.
- **CORR** ensures the trend matches ground truth.
- **SAGR** ensures the correct polarity of valence/arousal.
- **CCC** combines CORR + bias correction → most suited metric for “in-the-wild” deployment, since it balances association and offset.
In real-world use, high **CCC + SAGR** would be prioritized for robustness.

9. Conclusion

This work implemented and compared two modern CNN baselines → **EfficientNet-B0** and **MobileNetV3-Large** for the joint task of **facial expression recognition** and **valence-arousal estimation**. Both models were extended into a multi-task architecture with classification and regression heads, trained end-to-end with ImageNet-initialized weights.

The results highlight a clear **trade-off between efficiency and regression reliability**:

- **MobileNetV3-Large** achieved slightly higher **classification performance** (best ACC = 0.4858, F1 = 0.4798) while training **3.8× faster** (12.1 min vs 46.5 min). This makes it the stronger candidate when speed and deployment efficiency are the priority.

- **EfficientNet-B0** provided **better regression performance** on valence–arousal, with consistently higher **Pearson correlation** and **concordance correlation coefficient (CCC)** values, making it more suited for scenarios where **continuous affect estimation accuracy** is critical.

Qualitative analysis confirmed these trends: MobileNet tended to classify discrete categories more confidently but misclassified ambiguous cases into neutral, while EfficientNet produced more balanced outputs, aligning better with the continuous affect space.

In practical applications, the choice depends on the target system requirements:

- **Real-time systems** (e.g., mobile applications, live HCI) → **MobileNetV3-Large**.
- **Robust affect estimation in the wild** (e.g., psychological research, affective analytics) → **EfficientNet-B0**.

Ultimately, this comparison demonstrates that **model architecture choice is not only about accuracy, but also about balancing accuracy, generalization, and efficiency**. A hybrid approach (e.g., MobileNet for categorical pre-screening, EfficientNet for regression refinement) could combine the strengths of both.